



'ATHENA' A PROJECT BY AI WIZARDS



DETAILS OF THE TEAM MEMBERS



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THEME- AI IN EVERYDAY LIFE



PROBLEM STATEMENT

We make progress by standing on the shoulders of giants and that's true in any community whether it's science, psychology, history or art, we gain insights about the latest cutting edge technology by reading research papers written on that particular topic. Our goal is to develop an innovative AI chatbot to revolutionize research paper exploration and comprehension (Providing quizzes, diagrams to help users better understand the paper and predict future applications based on the research). Enhance knowledge exchange and democratize access to complex research for a wider audience.



PROPOSED SOLUTION

To build a Research Paper chatbot that can explain research papers, quiz the users, and predict applications based on the research paper, we need the following:

***Data Acquisition and Preprocessing:** Gathering a diverse dataset of research papers, including their titles, abstracts, and full text. Preprocessing the data by cleaning and removing irrelevant information.*

***Knowledge Extraction and Representation:** Utilize IBM Watson's Natural Language Understanding (NLU) to extract key entities, concepts, and relationships from the research papers. Construct a knowledge graph to represent the extracted information, connecting related concepts and papers.*



Question Answering and Explanation: Training a deep learning model, such as a transformer-based architecture, using TensorFlow, to answer questions about the research paper. Implement an explanation generation system that uses the knowledge graph to provide detailed explanations of the research papers.

Generating a Quiz: A generator that generates multiple-choice or open-ended questions based on the content of the research papers can be made. User responses can be evaluated using NLP techniques to provide feedback and 'score' their understanding of the research papers.

Applications Prediction: Employing neural networks and deep learning models to analyze the research paper's content to predict potential applications. Models are trained using labeled data or utilize transfer learning to use pre-trained models.



Chatbot Interface and Integration: Next step is to build a chatbot interface using IBM Watson Assistant, enabling users to interact with the system by inputting research papers and asking questions. Integrate the various components of the solution, including question answering, explanation generation, quiz generation, and application prediction.

Continuous Improvement using user feedback: Collecting user feedback to enhance the chatbot's performance, accuracy, and user experience can be used to Fine-tune the models, update the knowledge graph, and refine the algorithms.

By connecting IBM Watson, TensorFlow, neural networks, deep learning, and NLP techniques, this Research Paper AI chatbot can effectively explain research papers, quiz users, and predict applications. It provides researchers and users with a comprehensive understanding of research findings and their potential implications.



SOFTWARE/HARDWARE COMPONENTS NEEDED FOR THIS IMPLEMENTATION AND APPROXIMATE COST

- ❖ *NLP NLP libraries like NLTK*
- ❖ *IBM WATSON- Watson Natural Language Understanding (NLU)/TENSORFLOW (NEURAL NETWORKS AND DEEP LEARNING)*
- ❖ *PYTHON*
- ❖ *QISKIT(OPTIONAL)*
- ❖ *APPROXIMATE COST: 8-80lakh for starting from scratch but with pre available IBM cloud software the cost drops significantly.*



REFERENCES



- ❖ For NLP, Deep learning, chatbots, <https://proceedings.neurips.cc/>, <http://neuralnetworksanddeeplearning.com/>
- ❖ https://qiskit.org/ecosystem/machine-learning/tutorials/02_neural_network_classifier_and_regressor.html
- ❖ For research papers, <https://arxiv.org/> and James Chadwick's paper used to demonstrate the method, <https://web.mit.edu/22.54/resources/Chadwick.pdf>

THANK YOU